

Network diagram:

CPM (Critical path Method)

→ Network diagram depends upon the estimation. Every activity have two types of estimation

1) Deterministic → We use CPM method

2) Variable → We use PERT method → PERT (Project Evaluation and Review Techniques)

For CPM:

i) Draw network
ii) calculate the schedule for each activity - EST (Earliest Start Time)
LFT (Latest Finish Time)

EST: When can you start the activity

LFT: Max time that is allocate you need to finish

iii) Calculate EFT and LST
 $EFT = EST + \text{Time}$
→ (Earliest Finish time)

$LST = \text{Latest start time}$
 $= LFT - \text{Time}$

iv) Calculate Total Float
- Total float is always positive
- difference between start time
and finish time

Total float = $LST - EST$
or
 $LFT - EFT$

v) Identify Critical activities and
find Critical path.

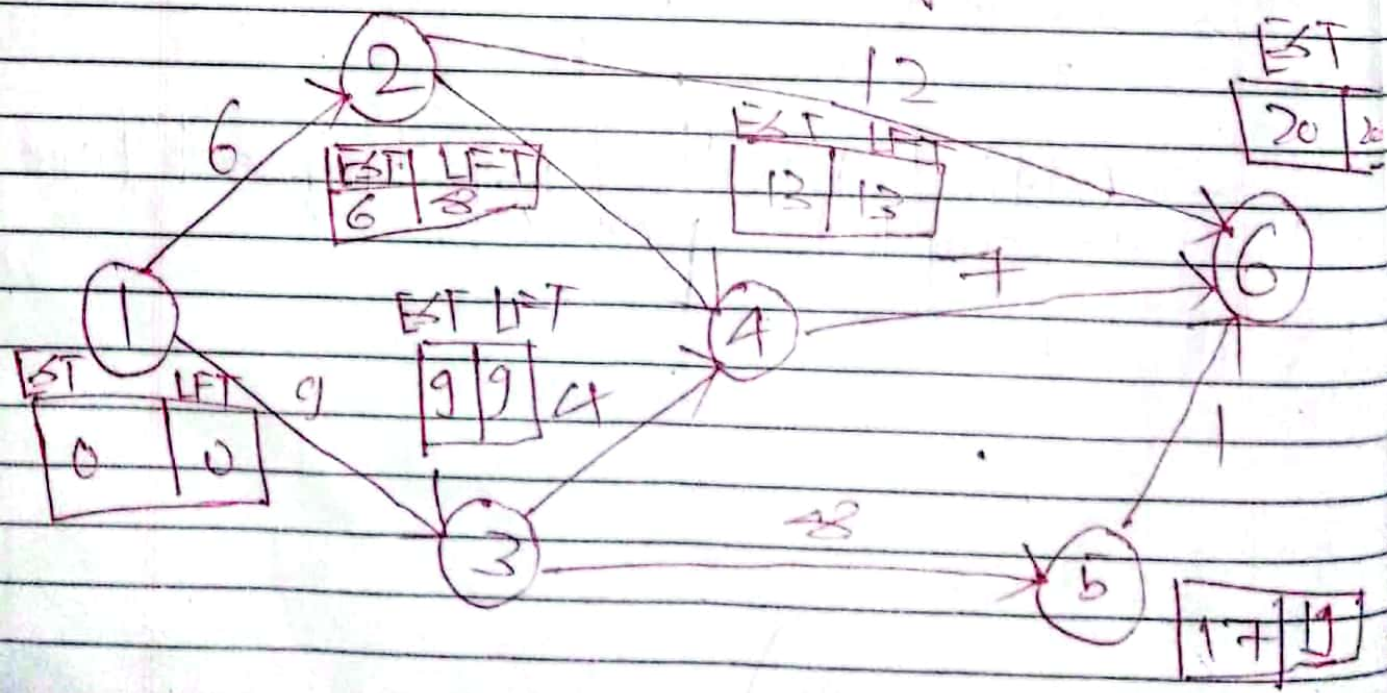
→ Critical method is longest time
to complete project

Q) Consider the following data and based on that find the Critical path and project completion time

Activity	Duration
1-2	6
1-3	9
2-4	3
3-4	4
3-5	8
2-6	12
4-6	7
5-6	1

⇒ i)

Draw Network Diagram



ii) Calculate EST and LFT
 EST (Highest Value)

iii) Calculate EFT and LST
 $EFT = EST + \text{Time}$
 $LST = LFT - \text{Time}$

iv) Total float
 $LST - EST$
 or
 $LFT - EFT$

3rd highest at 0

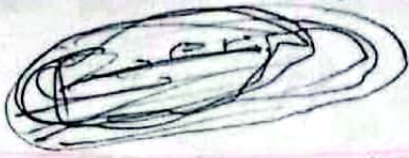
4th highest at 0

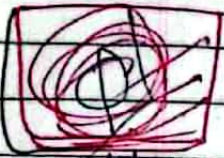
EST	LFT	EFT	LST	Float
0	8	6	2	2
0	9	9	0	0
6	13	9	10	4
9	13	13	9	0
9	14	17	11	2
6	20	18	18	2
13	20	20	13	0
17	20	18	19	2

3rd highest at 0

4th highest at 0

20


1-3 → 3-4 → 4-6 is Critical path


PERT (Project Evaluation and Review Technique)

→ PERT is used for various activities.

→ Optimistic Time Estimate (t_o)
(the minimum time required or best case)

→ Pessimistic Time Estimate (t_p)
(the maximum required or worst case)

→ Most likely time Estimate (t_m)
(Normal condition time)

1) Expected time or Average time

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

2) Variance

$$V = \left(\frac{t_p - t_o}{6} \right)^2$$

3) Standard deviation (critical activity Variance)

$$\sigma = \sqrt{V_1 + V_2 + V_3}$$

4) probability of completion of project

$$Z = \left(\frac{t - t_p}{\sigma} \right)$$

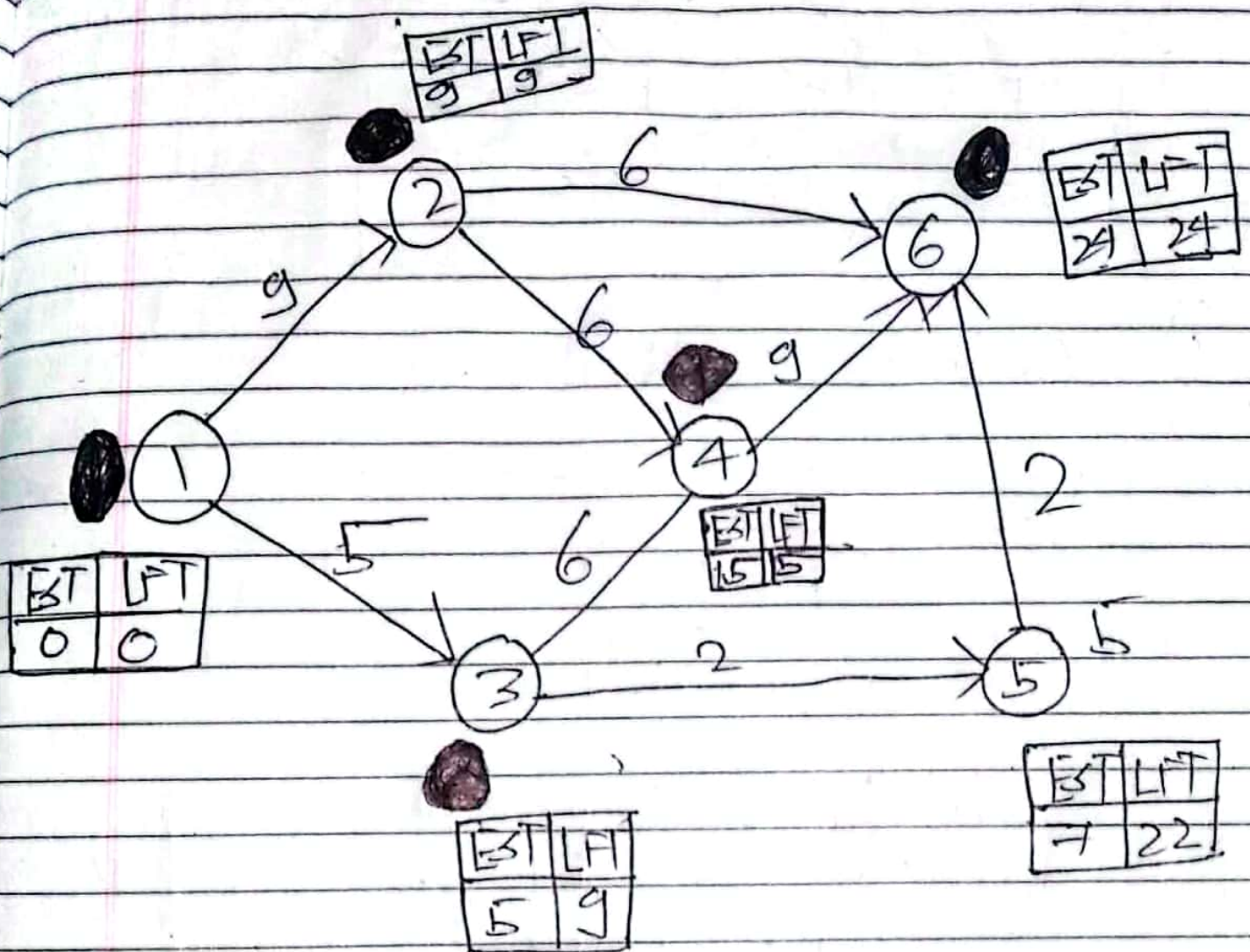
Q) For the given activities determine

- i) Critical path using PERT
- ii) Calculate Variance and standard deviation
- iii) Calculate probability of completing the project in 26 days

Activities	t_o	t_m	t_p	t_e
1-2	6	9	12	9
1-3	3	4	11	5
2-4	2	5	14	6
3-4	4	6	8	6
3-5	1	1.5	5	2
2-6	5	6	7	6
4-6	7	8	15	9
5-6	1	2	3	2

①

ET	LT
0	0



ET	LFT	EFT	LT	TF
0	9	9	0	0
9	9	15	4	4
5	15	11	9	0
5	22	7	9	4
9	24	15	20	15
15	24	24	18	9
7	24	9	15	0
			22	15

$$EFT = EST + 1 \text{ day}$$

$$LFT = LFT - 1 \text{ day}$$

$$TF = LST - EST$$

$$\text{or} \\ LFT - EFT$$

i) Critical path is

ii) $1-2 \rightarrow 2-4 \rightarrow 4-6$
Variance is

$$V_1 = 1$$

$$V_2 = 4$$

$$V_3 = 1.77$$

Standard deviation is

$$\sigma = \sqrt{1 + 4 + 1.77}$$

$$= \sqrt{6.77}$$

$$= 2.60$$

V	
1	
4	
1.77	

9.9 >

$$z = \left(\frac{t - t_{cr}}{6} \right)$$

$$= \left(\frac{26 - 24}{2.60} \right)$$

$$= 0.76$$

$$t = 26$$

$$t_{cr} = 24$$

Q) Draw network diagram and find EST, EFT, LST, LFT, TF (Slack) from following table.

Activity	Duration	Predecessor
A	3	-
B	6	-
C	2	-
D	2	A
E	1	B
F	3	C
G	7	B, D
H	3	E
I	4	B, F

or

Activity	Duration	Successor
A	3	D
B	6	E, G, I
C	2	F
D	2	G
E	1	H
F	3	I
G	7	-
H	3	-
I	4	-

Activity Successor में है तो
Start और End का

EST	LFT
0	0

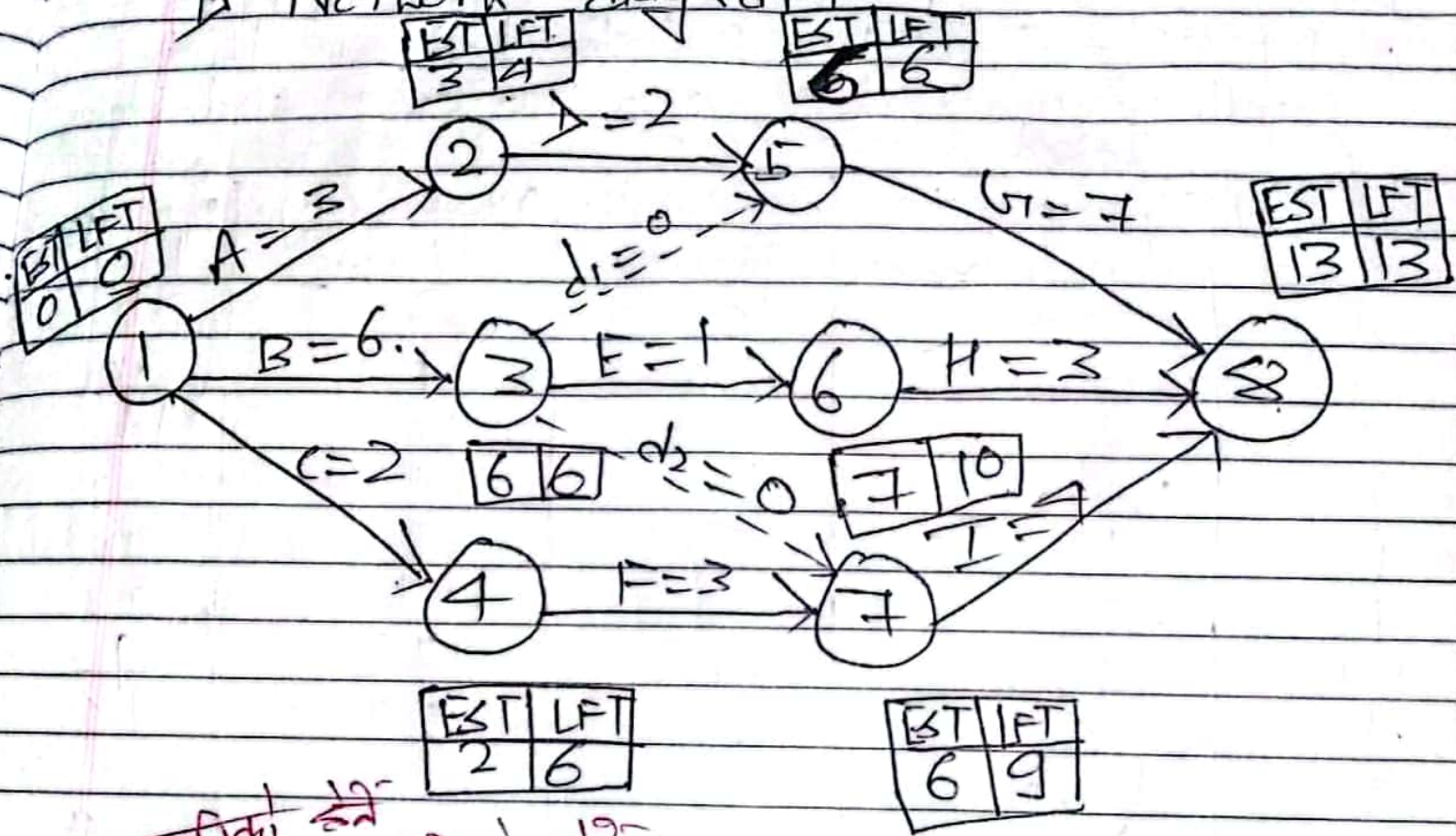
①

Activity	EST	LFT
A	0	0
B	0	6
C	0	2
D	3	5
E	6	7
F	2	4
G	5	12
H	7	8
I	6	10

Numbering left to right then top to bottom

→ solution:

1) Network diagram



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EST	LFT	EFT	LST	TF
0	4	4	0	
6	6	6	6	
0	6	6	0	
3	6	6	3	
6	10	10	6	
2	9	9	2	
6	13	13	6	
7	13	13	7	
6	13	13	6	

2018

$$FF = EFT - EST - \text{duration}$$

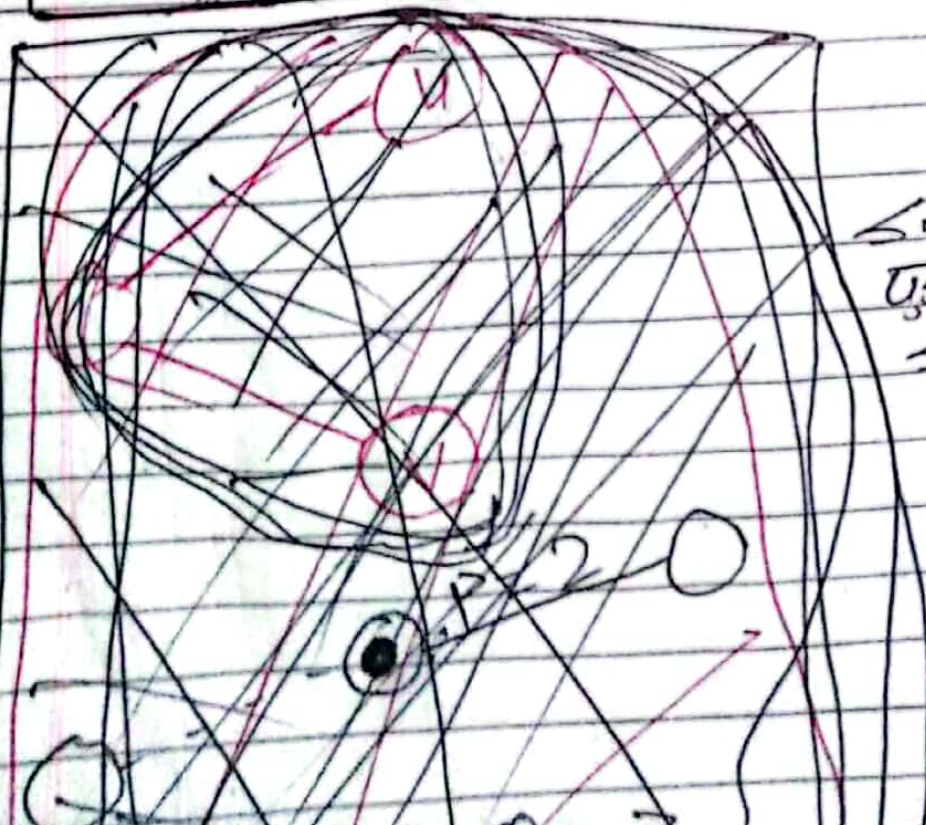
Draw network diagram and find EST, EFT, LST, LFT, TF (Slack) from the following table and Variance and standard deviation

Activity	Immediate Predecessor	optimistic time	Most likely time	pessimistic time
A	—	10	22	22
B	—	20	20	20
C	—	4	10	16
D	A	2	14	32
E	B, C	20	8	20
F	B, C	20	14	20
G	B, C	4	4	4
H	C	2	12	16
I	G, H	6	16	32
J	D, E	2	8	14

2017

Draw a network diagram for the given set of activities and identify critical path and critical activities. Also calculate project duration.

Activities	Duration	Immediate Successor
U → V	2	R
U → W	3	M
U → T	4	T, U
U → S	4	T, V
V → W	5	
W → S	3	
S → T	4	

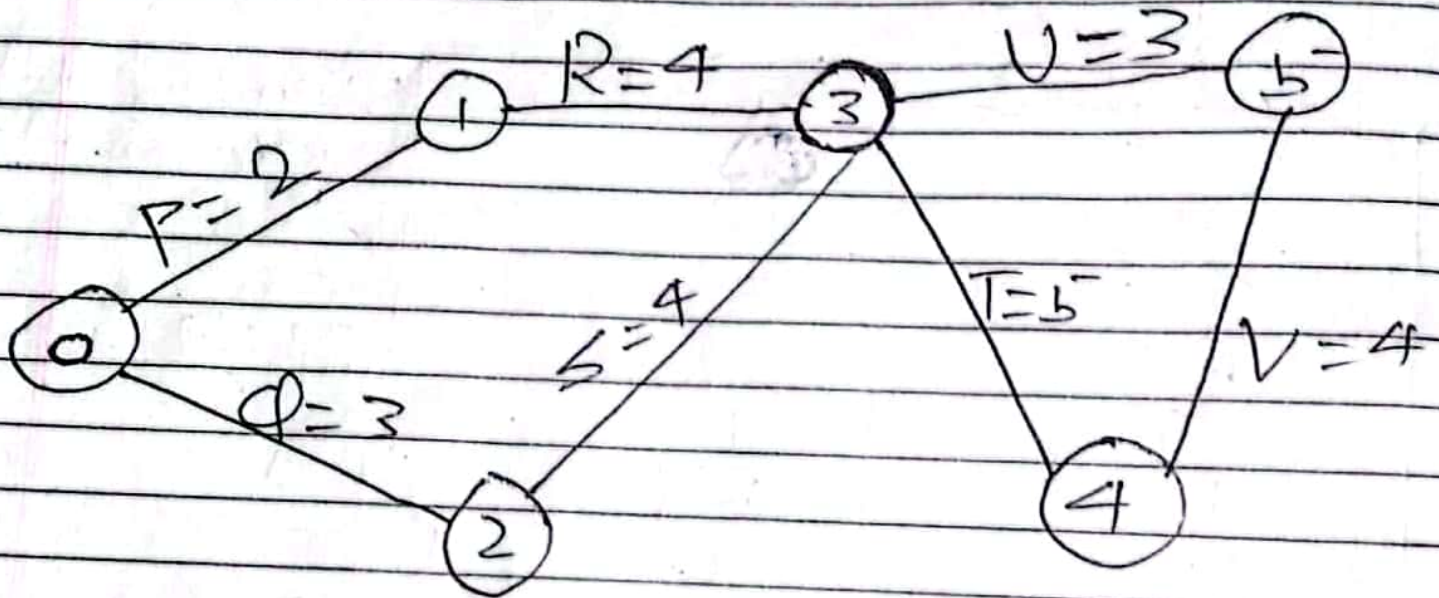


Successor एक ही होता है Common, Successor है और Common node

प्रश्न का उत्तर

या त सिद्ध वत बनाई या त लक्ष्य
सिद्ध वत बनाई ~~सफलतापूर्वक~~ सिद्ध
कर आसका है।

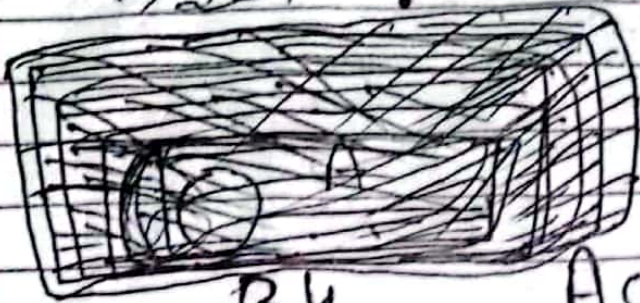
⇒ Network diagram:



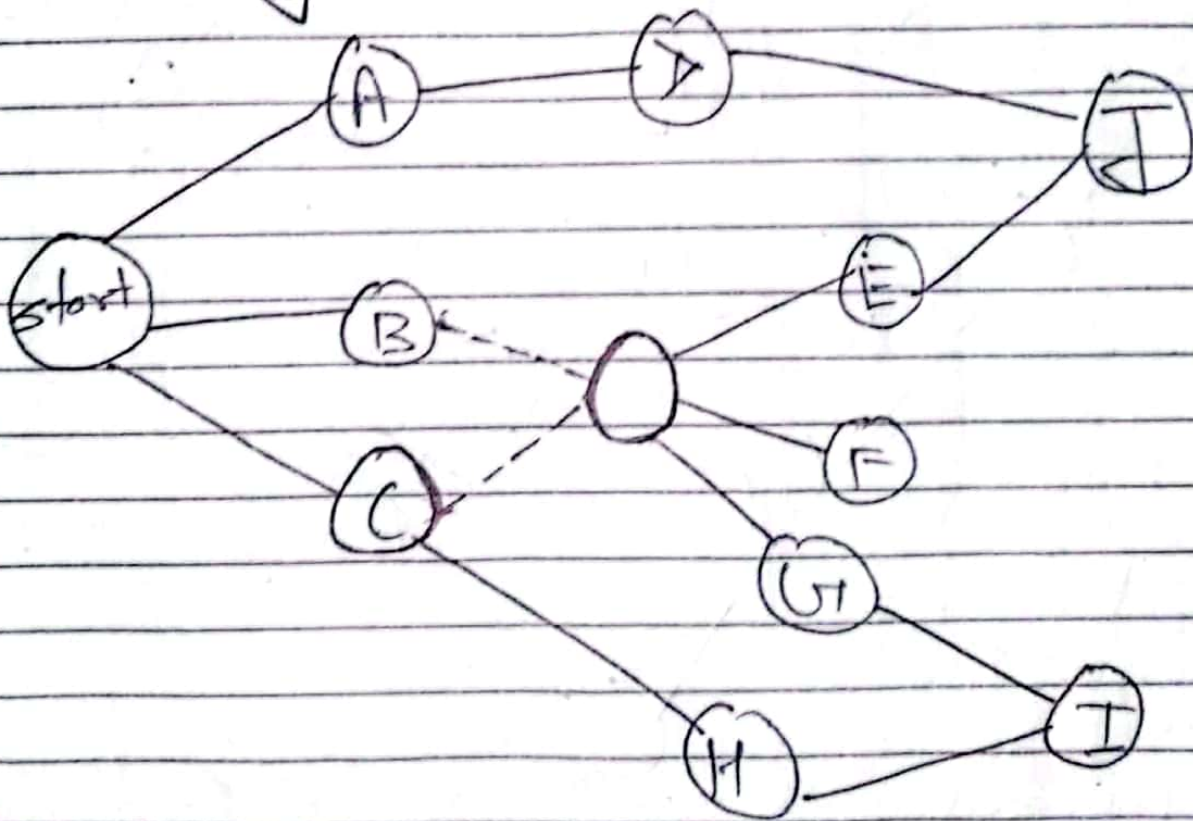
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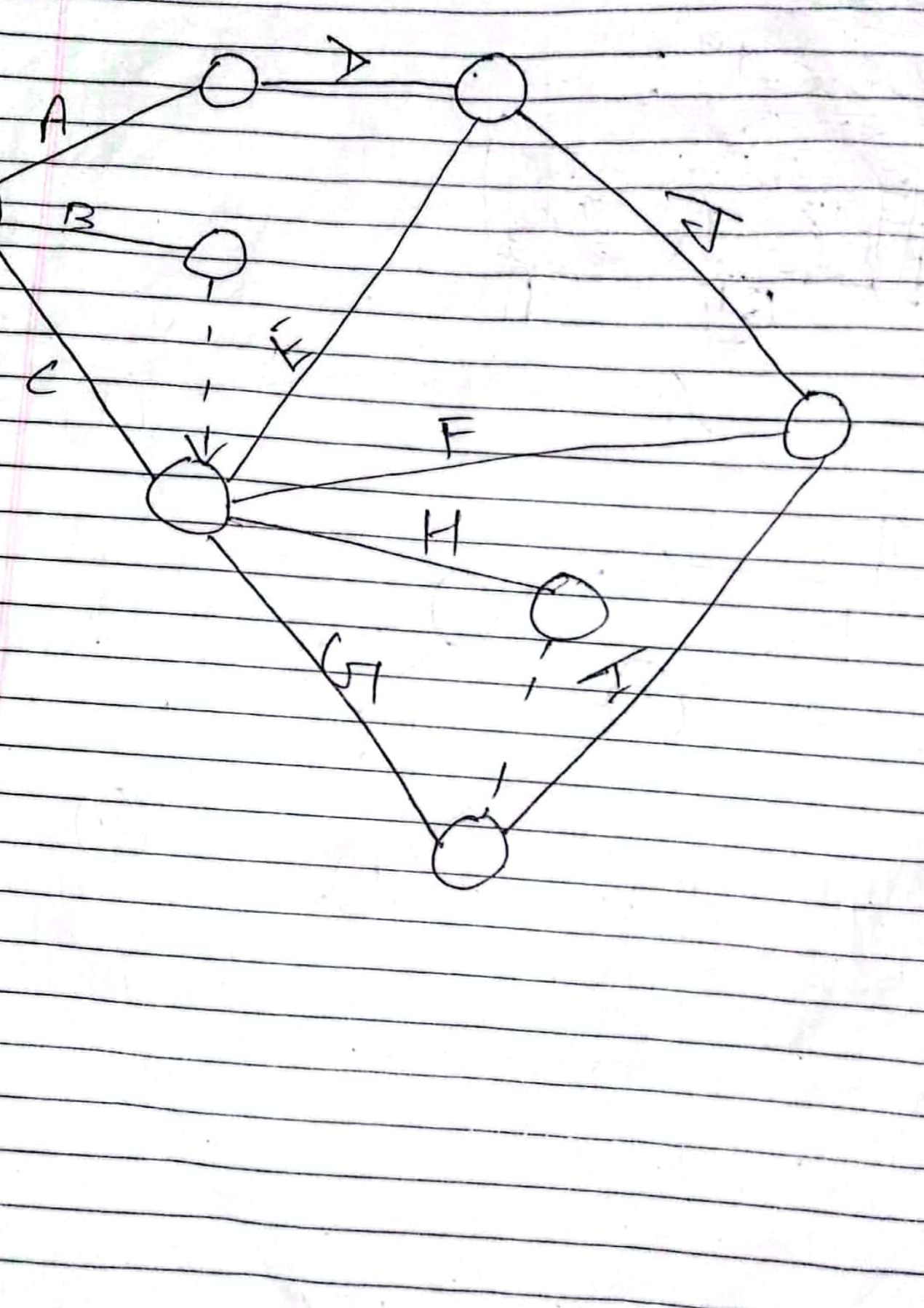
2018

> 50/11



By AON





Project Cost Management:

Terms	Formulas
CV = Cost Variance	$CV = EV - AC$
SV = Schedule Variance	$SV = EV - PV$
CPI = Cost Performance Index	$CPI = EV / AC$
SPI = Schedule Performance Index	$SPI = EV / PV$
EAC = Estimate at Completion	$EAC = BAC / CPI$
ETC = Estimate to Complete	$ETC = EAC - AC$
AC = Actual Cost	
BAC = Budget at Completion	
PV = Planned Value	
EV = Earned Value	

Q) Consider a project scheduled to complete within 4 months with the total estimated project cost Rs 4,00,000. At the end of 3 months Rs 2,00,000 was spent while work done was of cost Rs 1,00,000. With these dimensions, calculate CV, SV, CPI, SPI, EAC, ETC.

→ 2011:

Budget at Completion (BAC) = 400,000 L-

Cost Spent after 3 months

[Actual Cost (AC)] = 200,000 L-

Earned Value (EV) = 100,000 L-

Planned Value (PV) = 3,00,000 L-

$$CV = EV - AC$$

$$= 1,00,000 - 2,00,000$$

$$= -1,00,000 L-$$

The negative value means the project is over budget.

$$SV = EV - PV$$

$$= 1,00,000 - 3,00,000$$

$$= -2,00,000 L-$$

If SV is negative value then project is behind schedule.

$$CPI = \frac{EV}{AC} = \frac{1,00,000}{2,00,000} = 0.5$$

If CPI is less than 1 then project is over budget

$$SPI = \frac{EV}{PV} = \frac{1,00,000}{3,00,000} = 0.33$$

If SPI is less than 1 then project is behind schedule

$$EAC = \frac{BAC}{CPI} = \frac{4,00,000}{0.5} = 8,00,000$$

$$ETC = EAC - AC = 8,00,000 - 2,00,000 = 6,00,000$$

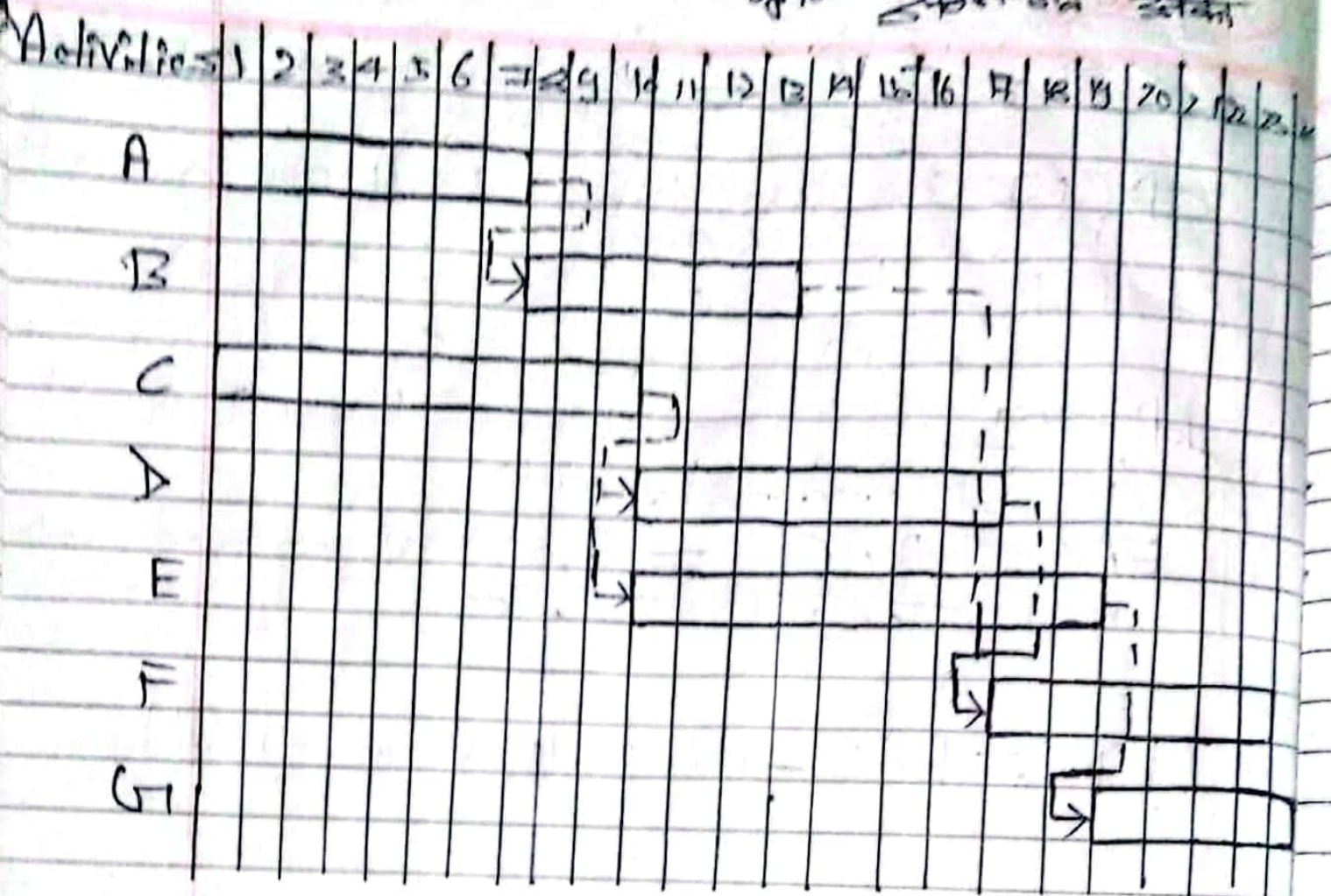
Network diagram
Linked bar chart
Gantt chart

2016

List out few tools used for modelling schedule. Draw a linked bar chart for the given set of activities.

Activities	Duration	Immediate Successor
A	7	B
B	6	F
C	10	D, E
D	7	F
E	9	G
F	6	-
G	5	-

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Prepare a probability vs impact matrix for your college's final year tour (at least four identifiable risks).

2021

Q Consider a project scheduled as following:

Activity	Predecessor	Duration	Cost/day	Total
A	-	2	300	600
B	A	3	400	1200
C	B	3	400	1200
D	B	2	200	400
E	D	3	100	300
		13		3700

After 7th day field report says something like below

Activity	Actual Complete (%)	Inurred Cost
A	100	600
B	100	1400
C	33	500
D	50	200
E	0	0
		2700

With these information calculate CV, SV, CPI, SPI, EAC, ETC

⇒ Solⁿ

BAC (Budget at Completion) = 3700
 Duration to Complete = 13 days
 Planned Value at the end of
 7th day = $\frac{100}{13} \times 7 = 53.8\%$

$$PV = 3700 \times \frac{53.8}{100} = 1992.3$$

$$\text{Actual Cost (AC)} = 2700$$

$$\text{Earn Value} = 600 + 1200 + (1200 \times 33\%) + (50\% \text{ of } 400)$$

$$= 600 + 1200 + \frac{33}{100} \times 1200 + \frac{50}{100} \times 400$$

$$(EV) = 2396$$